FORMATIVE ASSIGNMENT

Gaia

Project proposal

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Date created: September 26, 2026

1. Description of the concept

My ALU mission is to close the gap between underserved communities and the healthcare information and resources they need, starting in Rwanda. The mission sits inside the Healthcare access sector, One of the toughest problem areas in Africa: The continent carries a large share of the world's disease burden With far fewer health workers and health infrastructure than other regions. Rwanda has made real progress on Primary Health Care since the early 2000s, But still runs into a familiar wall: Care exists somewhere in the system, Yet patients and Health workers Often cannot see where.

2. Problem statement

Rwanda's health system runs in three levels, central referral hospitals, district and provincial hospitals, and health centers and posts, and referrals move between them whenever a case is too complex for the current facility. That referral decision is still made by hand. A study of referral letters across eleven hospitals in Kigali found transfer documents were often incomplete, especially clinical information and follow-up notes, weakening continuity of care between facilities (Kalume et al., 2022). That study only covers what happens after a referral decision is made, not the harder problem: how a nurse decides where to send a patient in the first place, usually by phone call, word of mouth, or by default to the nearest hospital, whether or not it can help.

This gap has grown more visible recently. From 2019 to September 2023, Babyl ran Rwanda's largest telemedicine platform, reaching 450 of the country's 510 health facilities and about two million patients, before its parent company's bankruptcy forced a sudden shutdown (Musange et al., 2026), showing how fast a centralized, foreign-owned digital health system can disappear. Even facilities that never used Babyl still face basic barriers: about a fifth of Rwandans face geographic barriers to healthcare despite primary care covering most service delivery (Shepard et al., 2023).

In short, rural and district clinics in Rwanda have no shared, real-time way to see bed availability, specialist presence, or supply levels at nearby facilities. Referral coordination still runs on phone calls and paper, tools that do not scale and leave no record of what a facility can actually offer at a given time. Gaia solves this with a simple, shared dashboard of resource status and a clear referral process, so a referral is based on what a facility can actually provide, not a guess.

3. Software Development model

This project will use an iterative waterfall model rather than a strict, single-pass waterfall or a fully agile approach. A pure waterfall model assumes all requirements are known from the start, which is not true here since I have not yet tested my assumptions with clinic staff. A fully agile approach does not fit either, since this assignment needs a fixed SRS document, and a backlog that shifts every sprint does not match a rubric that asks for a defined set of requirements up front. Iterative waterfall is the middle ground: the system still moves through the usual phases in order, requirements, design, build, testing, deployment, but the early phases can be revisited in short cycles as new information comes in. For Gaia, that looks like the steps below.

Requirements gathering. Write an initial set of requirements, then refine it through review with clinic-side stakeholders, or, for this assignment, through instructor and

peer feedback, before design starts.

System design. Design the dashboard, referral workflow, and access control, and check it against the requirements before building starts. A design flaw found here sends the process back to requirements.

Implementation. Build the core parts, login, resource status updates, referral creation and response, and reporting, in a set order, so each part is checked against its own requirements before the next begins.

Verification and testing. Test the requirements against the built system, and send any defects or gaps back to design or requirements instead of leaving them for later.

Deployment and review. Run a pilot, then hold a formal review where feedback from real use feeds back into a new round of requirements if needed.

This last step is the point of the model: a referral platform that gets requirements slightly wrong the first time is the difference between a tool clinic staff trust and one they quietly stop using.

4. Hypothesis

If Gaia is used across a small group of district clinics, the time between a referral decision and a patient arriving at a facility that can actually treat them will go down. Specifically, I expect referrals that end with a patient being redirected or turned away to fall by at least 20%, measured against a baseline drawn from facility referral logs or a short staff survey conducted in the month before the pilot begins, over a six-month pilot period across five to eight facilities. This is specific (redirected or declined referrals), measurable (a defined percentage reduction against a pre-pilot baseline), achievable (a small pilot, not a national rollout), realistic (only a web dashboard and basic devices, already common at facilities), and timely (a fixed six-month window).

5. References

Kalume, Z., Jansen, B., Nyssen, M., Cornelis, J., Verbeke, F., & Niyoyita, J. P. (2022). Assessment of formats and completeness of paper-based referral letters among urban hospitals in Rwanda: A retrospective baseline study. BMC Health Services Research, 22(1), Article 1436. https://doi.org/10.1186/s12913-022-08845-y

Musange, S. F., Absolomon, G., Umutoni, N., Umuhoza, S. M., Rubuga, F. K., Nsaba-Uwera, Y. D., Muhoza, P. S., Hitimana, R., Condo, J., & Humuza, J. (2026). Digital primary health in Rwanda: Qualitative study of user experiences and implementation lessons from Babyl's telemedicine platform. Journal of Medical Internet Research, 28, Article e84832. https://doi.org/10.2196/84832

Shepard, D. S., Halasa-Rappel, Y. A., Zeng, W., Rowlands, K. R., & Musange, S. F. (2023). Cost-effectiveness of expanding access to primary health care in rural Rwanda by adding laboratory-equipped health posts: A prospective, controlled study. American Journal of Tropical Medicine and Hygiene, 108(5), 1042-1051. https://doi.org/10.4269/ajtmh.22-0519

Software Requirements Specification (SRS)

Gaia

A Real-Time Referral and Resource Visibility Platform for Rwanda's Rural Health Centers

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Date created: September 26, 2026

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Revision History

Name	Date	Reason For Changes	Version
Ishimwe Axcel	09/26/2026	Initial draft for the document	1.0
Ishimwe Axcel	10/23/2026	UML Diagrams	2

1. Introduction

1.1 Purpose

This document specifies the software requirements for Gaia, version 1.0, a web-based platform that gives facility staff across Rwanda's district Health Network a shared, real-Time view of bed, specialists, and supply availability, and a structured workflow for creating and responding to patients referrals. This SRS Covers the full initial system: The resource dashboard, Referral workflow, Authentication and role-based access and District level reporting. It does not cover Downstream clinical systems such as electronic medical records Which are treated here only as future integration points rather than part of this release

1.2 Document Conventions

Functional requirements are labeled FR, grouped under a numbered feature (for example, FR 2 for the Referral management feature, with FR 2.1, FR 2.2, and so on for its individual requirements). Non-functional requirements are labeled NFR and numbered in order. Requirements are not ranked by priority in this document. Every requirement listed here is treated as necessary for version 1.0 release unless a later revision says otherwise

1.3 Intended Audience and Reading Suggestions

This SRS is written for the course instructor and peer reviewers grading this assignment As a working project document, it is also meant for anyone who later designs, builds, or evaluates Gaia: a course-based development team, District health office staff, or clinic staff checking whether the system matches their workflow. A reader who mainly wants to know what the system does should start with Section 2 (Overall Description) and Section 4 (System Features). A reader more interested in quality and constraints should read section 3 (External interface requirements) And Section 5 (Other non-functional requirements). Section 1 is worth reading first either way, since it sets the scope for everything after it.

1.4 Product Scope

Gaia is a lightweight tool for facility coordination, not a full hospital information system. Its goal is narrow on purpose: Replace the current phone-call-and-guesswork approach to referrals with a shared dashboard and a clear workflow, so a referral is based on what a facility can actually do right now, not an assumption. This supports the personal mission behind the project, Closing the gap between underserved communities and Healthcare Services meant to serve them, By making resources that already exist In Rwanda's Health System visible to the staff deciding where to send a patient. Version 1.0

targets a pilot five to eight District-level-facilities, Not a national rollout, In line with the iterative approach described in the project proposal.

1.5 System Overview

Gaia is a web-based platform that gives facility staff across Rwanda's district health network a shared, real-time view of bed, specialist, and supply availability at nearby facilities, paired with a structured workflow for creating, tracking, and responding to patient referrals. It runs as a standalone system (not a replacement for Rwanda's Health Management Information System) with a browser-based dashboard, a PostgreSQL backend, and SMS as a fallback channel for low-connectivity facilities. Three user classes interact with it: facility staff, who view resource status and initiate or respond to referrals; district health office administrators, who manage accounts and pull cross-facility reports; and, indirectly, patients, whose care is what the referral workflow is ultimately meant to speed up. Section 2 (Overall Description) covers the product's context, functions, and constraints in full detail.

2. Overall Description

2.1 Product Perspective

Gaia is a new, Standalone product, Not a follow-on to an existing system, Though it is built with an eye on what came before it. Babyl showed both the promise and the fragility of centralized digital Health platforms in Rwanda: Useful while it ran, but internally dependent on one foreign owned company whose bankruptcy shut it down overnight. Gaia takes a smaller, more distributed approach on purpose. It is a facility-to-facility coordination layer, Not a patient-facing consultation platform, So its usefulness does not depend on one company staying afloat. It works alongside existing referral practices, such as the paper transfer forms already in use, rather than replacing them, and it is not meant to replace Rwanda's Health Management information system. A future link to that system is a reasonable Next Step, but it is outside this initial scope.

2.2 Product Functions

- Give facility staff a live view of beds, Specialist, And Supply availability at nearby facilities.
- let facility staff create a referral request and have it delivered to the receiving facility in real time.
- Let receiving facility staff accept or decline referral based on current capacity.
- Let authorized staff keep their own facility's resources status current.
- Restrict what a user can see and do based on their role.
- Give District Health offices aggregate visibility to referral patterns and recurring resources Gaps.

2.3 User Classes and Characteristics

Three types of users are expected. Facility staff (Nurses and Community Health workers) Use the system more often and or the main audience for the resource dashboard and referral workflows. Their technical skill varies, so the interface has to stay simple no matter how much software experience someone has. Receiving facility Health staff overlap with this group but specifically handle referral responses(Accepting or declining incoming referrals), Which is more time-sensitive than routine dashboard browsing. District Health office administrators use the system less often but need the reporting functions, And need higher access to view data across facilities. Facility staff and receiving facility staff matter most to get right, since a workflow that fails them breaks the whole referral use case. District reporting is valuable but secondary to that core Loop

2.4 Operating Environment

Gaia is a web-based application, Open through a standard Browser on desktop computers, And on low cost Android devices, Which are more common at the facility level. It is expected to run over patchy, Low-bandwidth rural internet, With SMS has a fallback for the most urgent referral alerts. The backend is expected to run on Node.js With a postgreSQL database, matching the stock already used in related coursework for this project.

2.5 Design and Implementation Constraints

The system must follow Rwanda's data protection rules for patient-identifying information. It must stay usable on limited or unstable connectivity, which rules out any design that assumes a constant connection. Since this is a course project with a defined Pilot group, The first version favors a working core workflow over broad customization; Facility-specific settings are left for a later iteration, Not required for version 1.0. The system must also run on infrastructure a district health office can realistically maintain, which rules out designs that need specialized hosting.

2.6 User Documentation

A short onboarding guide will go out with the pilot, covering accounting setup, Updating the facilities resource status, and creating or responding to a referral. Given how much digital literacy varies among facility staff, The guide will lean on annotated screenshots rather than dense text, And will be available in both English and Kinyarwanda

2.7 Assumptions and Dependencies

The project assumes pilot facilities have at least one staff member who can operate dashboards on a basic smartphone or computer, and the facilities have some form of internet access, even if it comes and goes. It assumes District Health offices are willing to set up an administrator account for reporting. It also depends on facility staff actually keeping their resource status up to date; The system loses its value if status updates are neglected, Which is why NFR 6 (auditability) and FR 4.2 (stateless flagging) exist as safeguards rather than afterthoughts. If any of these assumptions turn out to be wrong during the pilot, the iterative model described in the project proposal allows the requirements to be revisited before more is built.

3. External Interface Requirements

3.1 User Interfaces

The main interface is a browser-based dashboard with two views: a facility view (the logged-in user's own facility status, which they can edit) and a network view (nearby facilities' status, used when starting a referral). Creating a referral follows a short form rather than a long questionnaire, since it is used under time pressure. Every screen keeps a header showing the logged-in user's facility and role, and error messages are written in plain language rather than technical terms, since the audience is clinical staff, not developers.

3.2 Hardware Interfaces

The system has no hardware interface of its own. It runs on standard consumer hardware already at facilities, desktop or laptop computers and Android phones or tablets, through a standard web browser and, where available, mobile data or Wi-Fi.

3.3 Software Interfaces

The system relies on a PostgreSQL database to store facility, user, and referral data, and on an SMS gateway service to send fallback referral alerts when there is no data connection. A future link to Rwanda's Health Management Information System is expected but is out of scope for version 1.0. The data model is kept simple in this first release so that link stays possible later without a full redesign.

3.4 Communications Interfaces

Client-server communication uses standard HTTPS to protect data in transit, since referral data includes sensitive patient information. Real-time referral alerts go over the same connection while a facility's

session is active, with SMS as a fallback when a facility cannot hold a stable connection. No email is planned for version 1.0, since email is not a reliable channel for the intended users.

4. Requirement Specification

Stakeholder Requirements Specification

Functional Requirements

Req ID	Requirements	Description
FR 1	**Facility Resource Dashboard**	View real-time resource status across nearby facilities.
FR 1.1	View bed availability	Allow facility staff to view real-time bed availability at nearby facilities.
FR 1.2	View specialist availability	Allow facility staff to see which specialists are on duty at nearby facilities.
FR 1.3	View supply stock levels	Allow facility staff to check current stock levels of critical medical supplies at nearby facilities.
FR 2	**Referral Management**	Create, transmit, and respond to referrals
FR 2.1	Initiate a referral	Allow facility staff to create a referral request specifying the patient's condition and the resource required.
FR 2.2	Receive referral notification	Notify the receiving facility of an incoming referral in real time.
FR 2.3	Accept or decline a referral	Allow receiving facility staff to accept or decline a referral request based on current capacity.
FR 2.4	Mark referral complete	Allow receiving facility staff to mark an accepted referral as complete once the patient has arrived and been treated. Only the receiving facility may do this, per the business rules in Section 6.7.
FR 2.5	Send SMS fallback alert	Automatically send an SMS notification of a new or updated referral to the receiving facility when the system cannot establish a data connection.
FR 2.6	View referral status	Allow facility staff to view the current status (pending, accepted, declined, completed, or cancelled) of a referral they initiated or received.

FR 2.7	Cancel a referral	Allow the referring facility to cancel a referral before it has been accepted by the receiving facility.
FR 3	**Authentication and Access Control**	Secure, role-based system access
FR 3.1	Facility staff login	Allow registered facility staff to log in to the system using a username and password.
FR 3.2	Role-based access	Restrict data visibility and available actions based on the logged-in user's role (community health worker, nurse, or district admin).
FR 3.3	Staff account creation	Allow a district health office administrator to create a facility staff account and assign it a role.
FR 3.4	logout	Allow a logged-in user to end their session.
FR 3.5	Password reset	Allow a user to reset a forgotten password through a verified, admin-assisted channel.
FR 4	**Resource Status Updates**	Keep facility data current
FR 4.1	Update resource status	Allow authorized staff to update their facility's bed, specialist, and supply status.
FR 4.2	Flag stale data	Automatically flag a facility's resource data as potentially outdated if it has not been updated within a set time threshold.
FR 5	**Reporting and Analytics**	Aggregate referral and resource-gap reporting
FR 5.1	Log referral outcomes	Log the outcome of each referral (accepted, declined, or completed) for later review.
FR 5.2	District-level reporting	Allow district health office staff to generate aggregate reports on referral patterns and recurring resource gaps.
FR 6	**Localization**	Let users work in their preferred language
FR 6.1	Switch languages	Allow a user to switch the interface between English and Kinyarwanda at any time.

5. Other Nonfunctional Requirements

5.1 Non-functional requirements summary

Requirement Type	Req ID	Description
Security	NFR 1	The system must authenticate every user through role-based login with securely hashed credentials before granting access.
Performance	NFR 2	The system must support at least 50 concurrent facility users without response times exceeding two seconds.
Reliability	NFR 3	The system must maintain at least 99% uptime, given that facilities depend on it for time-sensitive referral decisions.
Usability	NFR 4	The interface must be usable by staff with basic digital literacy: a first-time user must be able to complete a referral (initiate or respond) within 10 minutes using only the onboarding guide, and the interface must be available in both English and Kinyarwanda.
Connectivity	NFR 5	The system must remain functional on low-bandwidth rural connections, with critical referral alerts falling back to SMS where data connectivity is unavailable.
Auditabilty	NFR 6	All referral and status-update actions must be logged with a timestamp and user identifier to support auditing.
Data Protection	NFR 7	Patient-identifying information must be handled in accordance with Rwanda's data protection and privacy regulations.
Compatibility	NFR 8	The system must be accessible from standard desktop browsers and low-cost Android devices already in use at facilities.
Scalability	NFR 9	The system architecture must support onboarding additional districts without requiring a redesign of the core system.

6. Appendix

This section covers requirements that do not fit neatly into the sections above.

6.1 Database Requirements

- DB 1: All facility, user, resource status, and referral data must be stored in a relational database (PostgreSQL) with enforced links between records, so a referral can never point to a facility or user that does not exist.
- DB 2: The database must be backed up automatically at least once a day, and a backup must be restorable within four hours.
- DB 3: Referral and audit records must be kept for at least five years, and must never be deleted through the normal user interface.
- DB 4: Resource status history must be kept, not overwritten, so district offices can look at trends over time.

6.2 Internationalization Requirements

- I18N 1: The interface must be available in English and Kinyarwanda, and users must be able to switch languages at any time.
- I18N 2: All on-screen text must be stored in separate language files, not written directly into the code, so a language such as French can be added later without changing the system.
- I18N 3: Dates and times must use the Rwandan format (day/month/year) and Central Africa Time.

6.3 Legal and Regulatory Requirements

- LEG 1: The system must follow Rwanda's law on the protection of personal data and privacy, including rules on consent, storage, and sharing of patient information. This should be confirmed with legal or Ministry of Health guidance before any real pilot.
- LEG 2: Only the minimum patient information needed for a referral may be collected. Full medical records are outside the system's scope.
- LEG 3: Patient data should be stored in Rwanda where possible. If it must be stored elsewhere, that transfer must meet the legal requirements.
- LEG 4: A pilot in real facilities must have approval from the relevant district health office and the Ministry of Health before it starts.

6.4 Reuse Objectives

- REU 1: The system must be built in separate modules (dashboard, referrals, access control, reporting) so each one can be reused or replaced without rebuilding the others.
- REU 2: Facility, district, and language details must be settings, not fixed in the code, so Gaia can be set up in a new district or country with little rework.
- REU 3: The backend should expose a documented API so other systems, such as a future national health information system, can connect to it later.
- REU 4: Existing, well-supported open-source libraries should be used instead of custom code wherever possible, to keep the system easy to maintain.

6.5 Deployment and Maintenance Requirements

- DEP 1: The system must be deployable on a single standard server using common tools (Nginx, Node.js, PostgreSQL) that a district office or local IT team can maintain.
- DEP 2: Updates must be possible without more than 15 minutes of downtime, and outside normal clinic hours where possible.
- DEP 3: The system must record errors and failures in logs so problems can be found and fixed quickly.

6.6 Software Quality Attributes

Usability comes before configurability for this release, given how much digital literacy varies among facility staff and how time-sensitive referral decisions are. Reliability comes before feature breadth; a

smaller system that is always available is more useful here than a bigger one that sometimes goes down. Maintainability matters because the system is expected to grow past this pilot, so the code favors clear boundaries between the dashboard, referral workflow, and reporting functions over shortcuts that would be quicker to build but harder to extend later.

6.7 Business Rules

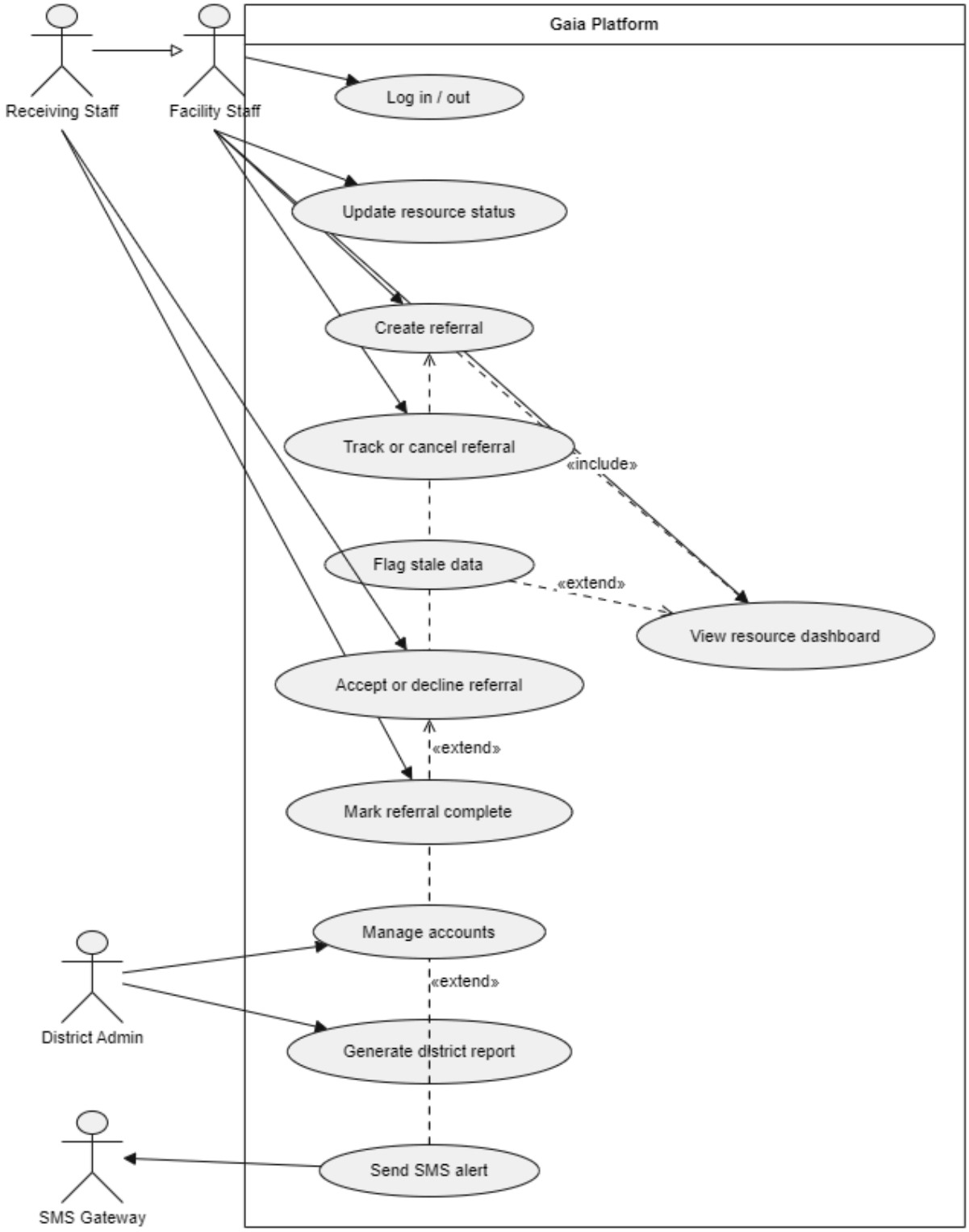
Only staff registered to a given facility may update that facility's resource status. Only district health office administrators may access cross-facility reports. A referral can only be marked complete by the receiving facility, not the referring one, since the receiving facility is the one that can actually confirm the patient arrived and was treated.

Appendix A: Glossary

Term	Definition
CHW	Community Health Worker; a frontline, often village-based, health worker who is typically the first point of contact for patients in rural areas.
District hospital	An intermediate-tier facility in Rwanda's health system that receives referrals from health centers and handles cases beyond primary-care scope.
Health center	A peripheral-tier primary care facility, usually the first formal point of care above the community level.
Referral	The process of transferring responsibility for a patient's care from one facility to another, typically because the required resource or specialty is unavailable at the originating facility.
Referral hospital	A central-tier, tertiary facility that provides specialized care beyond what district hospitals can offer.
Resource status	The current, facility-reported state of a specific resource (a bed category, a specialist's presence, or a supply item's stock level).
SRS	Software Requirements Specification; this document.

Appendix B: UML diagrams

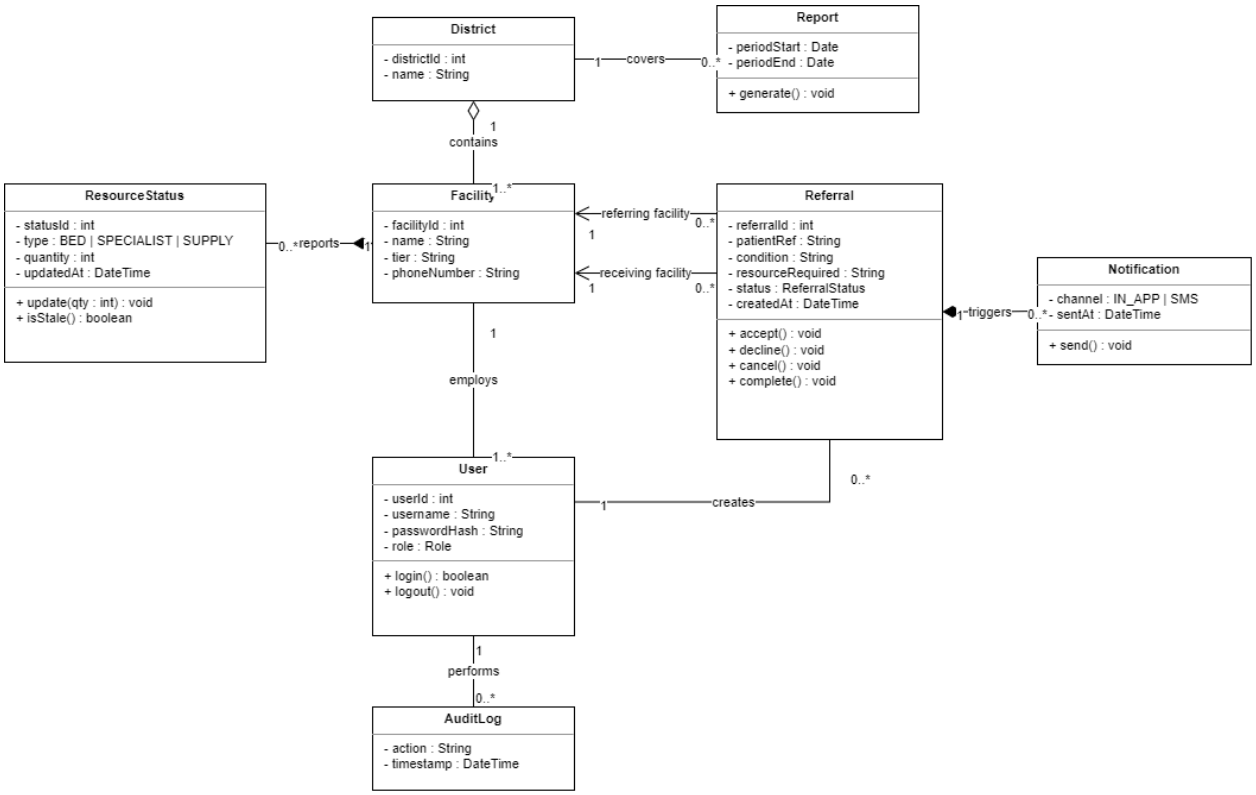
B.1 Use Case Diagram



This diagram shows who uses Gaia and what each person can do with it. There are four actors. Facility Staff (community health workers and nurses) view the resource dashboard, update their own facility's status, create referrals, and track or cancel the ones they sent. Receiving Staff are a type of Facility Staff, shown by the hollow arrow, so they can do everything above and also accept, decline, and mark referrals complete. The District Admin manages staff accounts (creating them and resetting passwords) and generates district reports. The SMS Gateway is an outside system that only takes part when a message has to be sent.

Three relationships connect the use cases. Creating a referral *includes* viewing the dashboard, because staff must check what a facility can offer before choosing where to send a patient. Flagging stale data *extends* the dashboard, since it only appears when a facility's data is old. Sending an SMS alert *extends* both creating and responding to a referral, because it only happens when there is no data connection. Every actor must log in before using any feature, and users can switch between English and Kinyarwanda from any screen.

B.2 Class Diagram

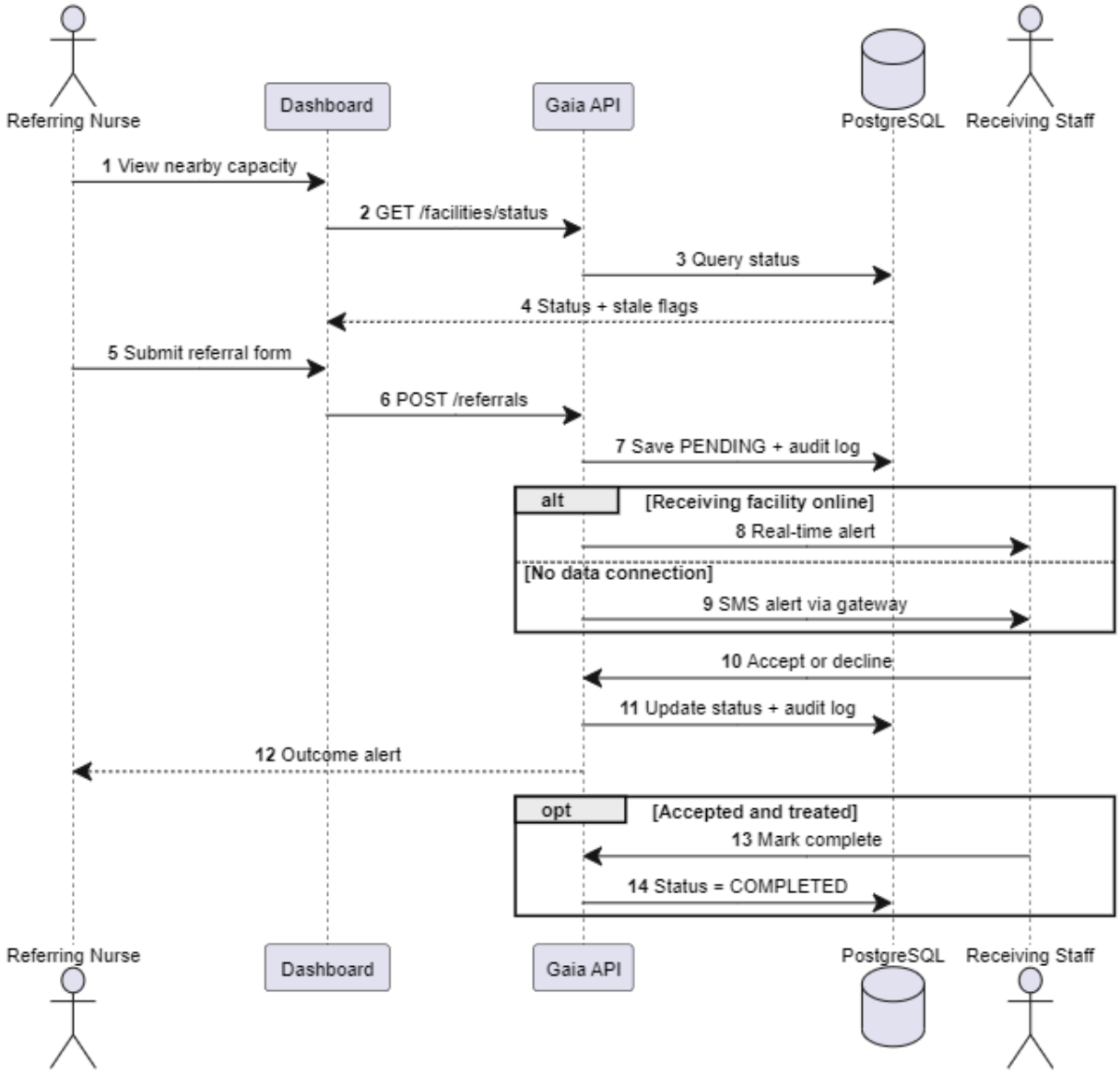


This diagram shows the main data Gaia stores and how the pieces connect. It matches the PostgreSQL database design in the appendix.

A **District** has one or more **Facilities**. Each Facility employs **Users** and owns its **ResourceStatus** records (beds, specialists, supplies). The filled diamond means these records belong to the facility and do not exist without it. A **Referral** connects two facilities, the one sending the patient and the one receiving, and is created by one User. It holds only the minimum patient information needed, such as the condition and the resource required. Each Referral can produce several **Notifications**, sent either in the app or by SMS. **AuditLog** records what each User does, and the Report is generated for a District by an admin.

Two values are enumerations. *Roles* can be community health workers, nurses, or district admin. *ReferralStatus* can be pending, accepted, declined, completed, or cancelled. The numbers on the lines (1, 0*.., 1..*) show how many objects can be linked. For example, one facility can have many referrals, but each referral has exactly one referring and one receiving facility.

B.3 Sequence Diagram



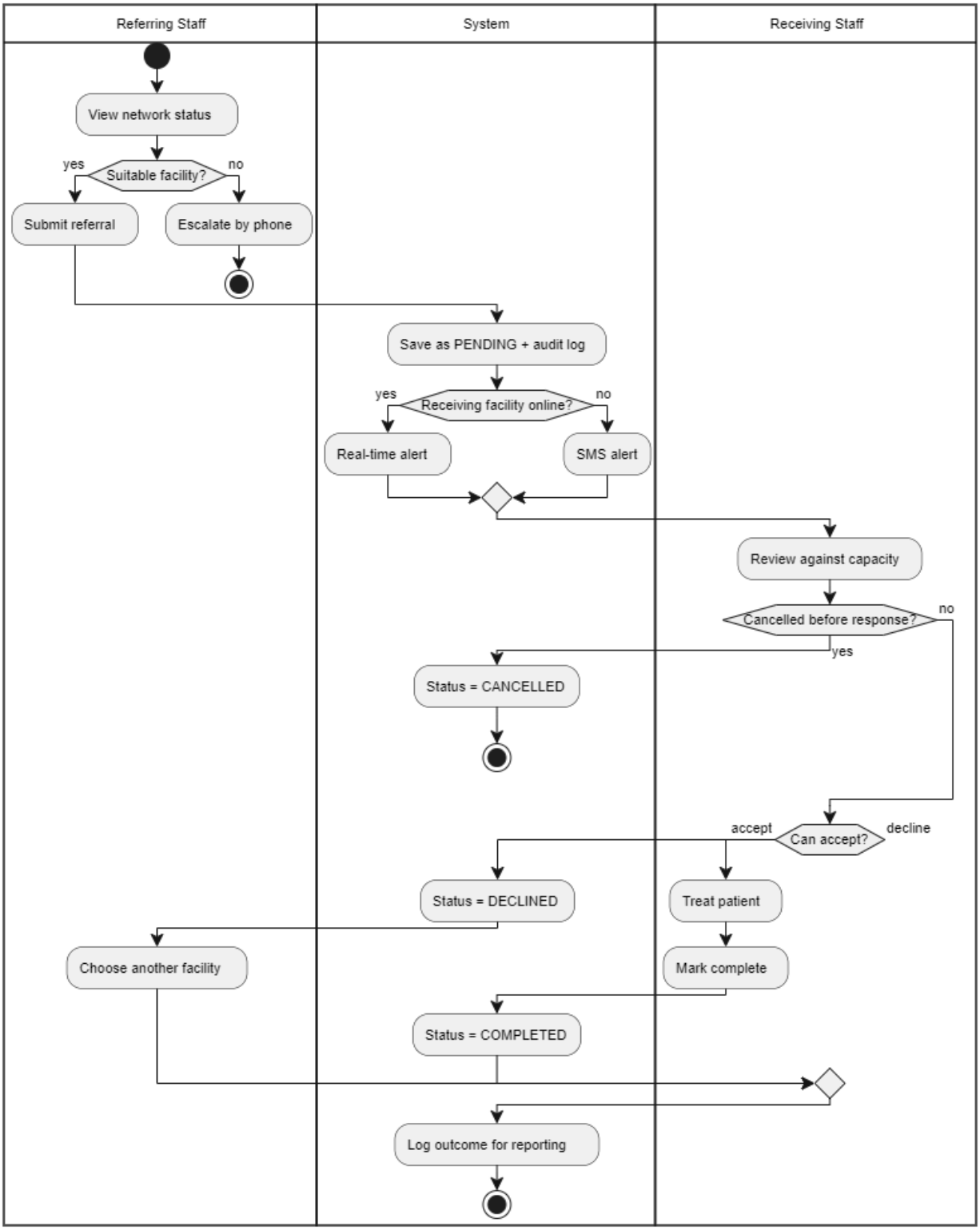
This diagram shows, in time order, what happens when a referral is created and answered. The participants are the Referring Nurse, the Dashboard (the screen the user sees), the Gaia API (the server), the PostgreSQL database, and the Receiving Staff. Messages are numbered, and arrows going down the page show later steps.

The nurse first asks to see nearby capacity. The dashboard requests it from the API, which reads the database and returns the resource status with any stale-data flags. The nurse then submits the referral form, which is the main input (patient condition and resource needed). The API saves it as pending, with an audit log entry, and alerts the receiving facility. The *alt* box shows two paths: a real-time alert if the

facility is online, or an SMS if it is not. In practice the API hands the alert to a notification service, which uses the SMS gateway when needed.

The receiving staff then accept or decline, the database updates, and the referring nurse receives the outcome as the output. The *opt* box shows an optional last step: if the patient was accepted and treated, the receiving staff mark the referral complete.

B.4 Activity Diagram



This diagram shows the whole referral workflow as a flow of actions and decisions. It has three swimlanes, one each for Referring Staff, the System, and Receiving Staff, so it is clear who does each step. The black dot is the start, the diamonds are decisions, and the circle with a ring is the end.

The process begins when the referring staff view the network status. If no facility has the resource, they escalate by phone and the process ends. If one does, they submit the referral. The system saves it as pending with an audit log entry, then sends a real-time alert or an SMS depending on whether the receiving facility is online. The receiving staff compare the referral with their current capacity. If it was cancelled before they responded, the status becomes cancelled and the flow ends. Otherwise they decide. On acceptance, the patient is treated, the referral is marked complete, and the system sets the status to completed. On decline, the status becomes declined and the referring staff choose another facility. Every path ends with the outcome being logged, which is what district reports are built from.